

Due date: 2/7/2025, end of the day.

1. Basic Python [18 points]

In this problem, you will need to follow the instructions in given .ipynb file (hw1.1 Basic Operation of Numpy and Matplotlib). The file includes a set of expected outcomes that this question aims to achieve. Your task is to develop your own code which should be capable of producing results similar to the provided expected outcomes.

/** Attention: 1) Please use a separate code block for EACH sub-problem. 2) Make sure to include sufficient discussions and insights for each sub-problem. 3) Use best practices of coding, e.g., including necessary comments to important statements and functions. **/

2. Polynomial Fitting [42 points]

In this problem, we write a program to estimate the parameters for an unknown polynomial using the `polyfit()` function of the numpy package.

- 1) Please plot the noisy data and the polynomial you found (in the same figure). You can use any value of m selected from 1, 2, 3, 4, 5, 6.
- 2) Plot MSE versus order m , for $m = 1, 2, 3, 4, 5, 6, 7, 8, 9$ respectively. Identify the best choice of m . Discuss how m impacts the performance in terms model accuracy and overfitting.
- 3) Initialize variable `noise_scale` to 150, 250, 350, 550, 750, and 950 respectively, re-run the algorithm, and plot the polynomials with the best m found in (2). Discuss the impact of increasing noise on the accuracy and reliability of the polynomial parameters. How does the model's performance degrade with increasing noise? [You need to plot a figure like in (1) for EACH choice of noise scale.]
- 4) Change variable `number_of_samples` to 100, 80, 40, 30, 20, and 10 respectively, re-run the algorithm, and plot the polynomials with the m found in (ii). Discuss the impact of the number of samples on the accuracy of the returned parameters. [You need to plot a figure like in (i) for EACH choice of number_of_samples.]

Please use the following code at the beginning of your program to generate the data.

```
import matplotlib.pyplot as plt

plt.style.use('seaborn-whitegrid')
import numpy as np

Your_Stevens_ID =          # use your real Stevens ID here
noise_scale = 900          # In question 3) modify this number
number_of_samples = 70     # In question 4) modify this number
noise_scale += Your_Stevens_ID % 70
CWID = Your_Stevens_ID % 4 + 3

x = 30 * (np.random.rand(number_of_samples, 1) - 0.5)
if CWID == 3:
    y = 6 * x + 7 * x**2 + 6 * x**3 + 10*noise_scale*np.random.randn(number_of_samples, 1)
elif CWID == 4:
    y = 6 * x + 7 * x**2 + 5 * x**4 + 100*noise_scale*np.random.randn(number_of_samples, 1)
elif CWID == 5:
    y = 6 * x + 8 * x**2 + 4 * x**5 + 1000*noise_scale*np.random.randn(number_of_samples, 1)
else:
    y = 6 * x + 7 * x**2 + 3 * x**6 + 10000*noise_scale*np.random.randn(number_of_samples, 1)

plt.plot(x,y,'ro')
```